Replicating Scene-Graph Memory for Robotic Imitation Learning

Expanding Spatial and Temporal Context for Robotic Imitation Learning With Scene Graphs

1. Introduction and project goals

I want to understand how robots remember objects they can no longer see. I plan to replicate the scene-graph memory method proposed by Qian et al. [1] and check whether I can reproduce its main findings. This project connects my interest in robotic memory with the transformer concepts in this course, without proposing a new architecture.

2. Research hypotheses and machine learning method

A robot's camera only captures part of its surroundings, so objects outside its view may still matter later. I will test two hypotheses: **H1:** Remembering previously observed objects improves task success over using only visible objects. **H2:** Removing 3D positions reduces success on tasks that depend on remembered locations.

Following the paper, each object will have visual features, a bounding box when visible, and a remembered 3D position. A transformer will process these object tokens alongside a scene-level token. Its output will guide a diffusion policy trained on expert demonstrations through noise prediction [2].

3. Related work

Alongside the main paper [1], I will study Diffusion Policy [2] and DINOv2 [3] to understand action learning and object features. The authors' repository [4] will be my implementation starting point.

4. Dataset, implementation, and evaluation plan

I will begin with one original task, using the authors' demonstration data. The repository provides 10 sample episodes per task [4], which I will use to check preprocessing before confirming access to the full dataset and evaluation setup. Any missing materials or changes to the original method will be documented.

I will compare the original policy with a visible-objects baseline and a version without 3D positions, keeping data and evaluation conditions consistent. I will report task and subtask success across repeated runs, with trial counts and confidence intervals. Offline prediction results will be distinguished from tasks actually completed in the simulator or robot setup.

5. Expected learning and conclusion

I hope to understand when scene memory helps and how the transformer uses it. My final report will include code, experimental comparisons, and an explanation of which findings I reproduced, where results differed, and what may explain those differences.

References

[1] J. Qian et al. (2026). *Expanding Spatial and Temporal Context for Robotic Imitation Learning With Scene Graphs.* arXiv:2606.01072.

[2] C. Chi et al. (2023). *Diffusion Policy: Visuomotor Policy Learning via Action Diffusion.* arXiv:2303.04137.

[3] M. Oquab et al. (2023). *DINOv2: Learning Robust Visual Features without Supervision.* arXiv:2304.07193.

[4] J. Qian et al. *scene-graph-robot: Official project repository.* GitHub (accessed September 13, 2026).